

Course Code	Course Title	L	T	P	C
BACSE344	Cloud Infrastructure and Architecture	3	1	0	4
Pre-requisite	NIL	Syllabus Version			
		1.0			
Course Objectives					
1. To evaluate cloud models and strategies to meet business goals. 2. To examine strategies for resilient infrastructure using core and containerized mechanisms. 3. To impart knowledge of cloud governance frameworks that ensure secure, cost-effective services, with emphasis on robust Service Level Agreements (SLAs).					
Course Outcomes					
1. Understand cloud service and deployment models to align with strategic business goals. 2. Analyze cloud infrastructure mechanisms including virtualization, storage, network perimeter, CloudWatch monitoring and replication. 3. Interpret the interactions between Docker packaging and Kubernetes orchestration in achieving scalable operation. 4. Develop a pattern-driven solution outline for dynamic scalability and cloud bursting under variable load. 5. Apply security and cost-management principles to cloud services using SLA, billing, resource-management mechanisms.					
Module:1	Fundamentals of Cloud Architecture	8 hours			
Business Drivers and Technology Innovations: Grid Computing - Serverless - Roles and Boundaries - Goals and Benefits - Cloud Characteristics - Infrastructure as a Service (IaaS) - Platform as a Service (PaaS) - Software as a Service (SaaS) Service Models - Cloud Deployment Models - National Institute of Standards and Technology (NIST) Reference Architectures - Web Services and Representational State Transfer (REST), Application Programming Interfaces (APIs).					

Module:2	Virtualization and Cloud Infrastructure	9 hours
Data Center Technology - Service Agents and Middleware - Logical Network Perimeter Mechanism - Fundamentals of Virtualization - Virtualization Technology - Virtual Server - Hypervisor - Cloud Storage Device - Cloud Usage Monitor: CloudWatch - Resource Replication.		
Module:3	Containerization and Orchestration	9 hours
Automated Scaling Listener - Load Balancer - Failover System - Resource Cluster - Containerization - Container vs. Virtual Machine - Common Container Technologies - Docker Server and Components - Kubernetes Node - Pod - Kubelet - Kubernetes Cluster and Control Plane.		
Module:4	Architectural Design, Patterns and Composition	9 hours
Workload Distribution Architecture - Resource Pooling - Dynamic Scalability - Elastic Resource Capacity - Service Load Balancing - Cloud Bursting - Load-Balanced Virtual Server Instances - Zero Downtime and Non-Disruptive Relocation - Virtual Private Cloud (VPC) and Multi Cloud Architectures - Specialized Cloud Architecture: Edge and Fog Computing Architectures		
Module:5	Cloud Security and Management	8 hours
Threat Agents - Common Threats - Hashing - Single Sign On - Multifactor Authentication (MFA) - Hardened Images - Trusted Platform Module (TPM) - Cloud Management Mechanism: Remote Administration System - Resource Management System - SLA Management System - Billing Management System - Compliance and regulatory standards: Health Insurance Portability and Accountability Act (HIPAA) - General Data Protection Regulation (GDPR)		
Module:6	Contemporary Issues	2 hours
Total Lecture Hours:		45 hours
Text Book(s)		
1. Thomas Erl and Eric Barcelo Monroy, "Cloud Computing: Concepts, Technology, Security, and Architecture", Pearson, 2nd Edition, 2023 2. Kyle Brown, Bobby Woolf and Joseph Yoder, "Cloud Application Architecture Patterns", O Reilly Media, 1st Edition, 2024		
Reference Books		
1. Rajkumar Buyya, Christian Vecchiola and S. Thamarai Selvi, "Mastering Cloud Computing: Foundations and Applications Programming", Morgan Kaufmann, 1st Edition, 2013		

2. Justin Garrison and Kris Nova, "Cloud Native Infrastructure: Patterns for Scalable Infrastructure and Applications in a Dynamic Environment", O Reilly Media, 1st Edition, 2017 3. Fang Liu, Jin Tong, Jian Mao, Robert Bohn, John Messina, Lee Badger and Dawn Leaf, "NIST Cloud Computing Reference Architecture", NIST Special Publication, 1st Edition, 2011	
Mode of Evaluation: Continuous Assessment Test, Quiz, Final Assessment Test, Course Project, Case Study	
Recommended by Board of Studies :	16-10-2025
Approved by Academic Council : No. 80	20-11-2025